

AI SUPPORT TICKET TRIAGE

Multi-Step Prompt Workflow for Extracting, Reasoning and Formatting Support Tickets

STUDENT DETAILS

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Assessment	Mini Assessment – Prompt Workflow Design

01 | Concept

AI Support Ticket Triage converts an unstructured customer support ticket into a structured result containing key entities, an urgency level and a concise summary. The workflow breaks the task into smaller connected prompts so that each stage has a clear responsibility.

02 | Objective

The workflow uses sequential prompts to extract important information, reason about urgency and produce a machine-readable final response. Each stage uses the output generated by the previous stage.

03 | Workflow Overview

INPUT	→	1 Extract Few-shot	→	2 Reason CoT	→	3 Format JSON	→	OUTPUT
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04 | Workflow Steps

1. Extract Key Entities – Few-shot

Extract product/service, problem, duration, customer impact and business impact from the support ticket using examples to anchor the expected output format.

Output: Extracted entity list.

2. Determine Urgency – Chain-of-Thought

Use the actual entities produced by Step 1. Reason about severity, duration, customer impact and business impact, then classify urgency as low, medium or high.

Output: Urgency level with reasoning.

3. Format Final Result – JSON Mode

Combine Step 1 and Step 2 into a strict JSON object with exactly three keys: entities, urgency and summary.

Output: Valid parseable JSON.

05 | Prompt Workflow

Step 1 – Few-shot Entity Extraction

You are a support-ticket entity extraction assistant.

Task: Extract the key entities from the support ticket.

Return the entities as a bullet list containing only important information such as product or service, problem, customer impact, error details and time-related information.

Examples:

Support ticket: "My laptop has been showing a blue screen error since yesterday and I cannot access my work files."

Entities: Product: laptop; Problem: blue screen error; Customer impact: cannot access work files; Time: since yesterday.

Support ticket: "I was charged twice for my monthly subscription."

Entities: Service: monthly subscription; Problem: duplicate charge; Customer impact: charged twice.

Support ticket: {SUPPORT_TICKET}

Extract the key entities using the same format as the examples.

Step 2 – CoT Urgency Reasoning

You are a senior customer-support triage specialist.

Task: Determine the urgency level using the entities extracted in Step 1.

Possible urgency levels: low, medium, high.

Reason step by step using: (1) severity, (2) duration, (3) customer impact, and (4) business impact. Then give the final urgency level.

Step 1 extracted entities:

{STEP_1_OUTPUT}

Determine the urgency based only on the information provided above.

Step 3 – JSON Formatting

You are a support-ticket reporting assistant.

Using the outputs from Step 1 and Step 2, create the final support-ticket report.

Return ONLY valid JSON.

The JSON must contain exactly these three keys:

{"entities": [], "urgency": "", "summary": ""}

Rules: entities must contain the extracted entities; urgency must be low, medium or high; summary must be one concise sentence; do not add additional keys; do not use Markdown or explanations outside JSON.

Step 1 output: {STEP_1_OUTPUT}

Step 2 output: {STEP_2_OUTPUT}

06 | Prompt Chaining

Step 1 produces the entity list; Step 2 uses that actual output to determine urgency; Step 3 uses both previous outputs to create the final JSON. This demonstrates genuine multi-step chaining rather than three independent prompts.

STAGE	INPUT / CONTEXT	TECHNIQUE	OUTPUT
1	Support ticket	Few-shot	Key entities
2	Step 1 entities	CoT reasoning	Urgency level
3	Step 1 + Step 2	JSON mode	Final JSON

07 | Demonstration Input

"My company's payment portal has been down for the last 3 hours. I am unable to process customer payments, and we are losing sales."

08 | Intermediate Outputs

Step 1 – Extracted Entities

- Service: company payment portal
- Problem: portal is down
- Duration: last 3 hours
- Customer impact: unable to process customer payments
- Business impact: losing sales

Step 2 – Urgency

The issue prevents payment processing and is already causing business losses. The service has been unavailable for three hours, making the incident time-sensitive.

Final urgency: high

09 | Expected Final Output

```
{
  "entities": [
    "Service: company payment portal",
    "Problem: portal is down",
    "Duration: last 3 hours",
    "Customer impact: unable to process customer payments",
    "Business impact: losing sales"
  ],
  "urgency": "high",
  "summary": "The company payment portal has been unavailable for three hours, preventing customer payments and causing sales losses."
}
```

10 | Evaluation Checklist

Requirement	Status
Three chained workflow steps	Completed
Step 1 uses few-shot prompting	Completed
Step 2 uses CoT reasoning	Completed
Step 2 conditions on Step 1 output	Completed
Step 3 uses JSON mode / structured output	Completed

Final JSON has entities, urgency and summary	Completed
End-to-end example included	Completed
No coding required for workflow design	Completed

11 | Conclusion

The AI Support Ticket Triage workflow demonstrates prompt chaining by separating extraction, reasoning and formatting into connected stages. The result is a clear, reusable and structured workflow that can be tested directly in an AI interface without building software.